An established traffic backbone for an elementary VSL task

Literature comparison and formed modeling backbone — source hierarchy, 22 October 2026

1. What a suitable backbone must provide

Can an established variable speed limit (VSL) model support the information, control and gradient questions with few traffic-model changes? Consider an ending lane E, its adjacent continuing lane A and farther lane B. Initial states can share the same E and combined A+B traffic at every road position while differing in A/B loading. We ask whether that difference changes merge service, useful common-speed advice and its sensitivity. A suitable backbone must represent the distinction, conserve traffic, respect receiving space and count every initial unit's time until exit. Simplicity concerns the traffic rules we must author and justify, not citation count.

2. What the closest published models offer

Aggregate VSL models support questions about traffic location or history; lane models additionally resolve A/B loading. The inspected starting points offer the following tradeoff.

Table 1. Published starting points and their fit to the question

Published model	What is already supplied	Direct finding for this review
Jin and Jin [1]	Spatial and link-average traffic; VSL entry control, capacity drop and upstream holding queues.	Simpler if the question concerns *where* traffic lies along the road. It changes the A/B information question.
Wei, Han and Wang [2]	Link traffic retaining entry/speed history, interacting speed cohorts and capacity drop.	A strong aggregate alternative, but no explicit A/B loading. Native advice acts on entering traffic.
Nagalur Subraveti et al. [3]	Successive lane-cell densities, lane-change incentives and a lane-drop case; Han congestion lineage [4].	Selected lane backbone. Adopt Han's VSL sending restriction [5] and explicitly replace movement realization.
Carey et al. [6]	Multilane cell transmission, mandatory movements and receiving-allocation alternatives.	Supports conservative service and forward fallback. More VSL/lane-end task closure would be needed as the main backbone.

Few-dependency construction. Use Subraveti's lane framework as the backbone and follow its adoption of Han's congestion approach, completing and adapting that formulation for the VSL task. Replace movement realization with one Carey-led adaptation; Wright supports its allocation rule. The thesis clarifies the backbone. Thus *one principal framework with inherited budgets and one adapted movement component* carries the traffic model. Our scenario and composition remain explicit. This concentrates independent additions within the inspected literature; global minimality is not proved. Jin–Jin is simpler if the information question changes to spatial traffic differences.

Source hierarchy and relationship key. Columns show architectural roles; colors credit rules and references, not ownership of symbols.

Principal framework	**Inherited budget lineage**	**Movement adaptation**
Subraveti article [3]	Han 2016 [4] (adopted in [3])	Carey et al. [6] (principal)
Subraveti thesis [7] (clarification)	Han 2017 [5] (VSL extension)	Wright et al. [8] (supporting)

Comparison only: Jin–Jin [1]; Wei et al. [2]. This study: scenario, composition and task (**D**).

B: source form retained/re-expressed; **A**: adapted/composed from the colored source; **D**: defined/derived here, including member choices (not a novelty claim); S: standard algebra, conservation or calculus.

3. The formed modeling backbone

3.1. Lane-cell traffic state. The lane-density architecture (B) forms a *lane-resolved cell-transmission-type (CTM) model* with inherited Han budgets. Physical cells retain density and finite storage throughout the road; there is no separate link transmission model (LTM) link. E can change into A at upstream cell boundaries; traffic in the final E cell must merge at the exit. Unserved mass stays in its current cell, so congestion can extend upstream. The causal chain is:

Lane stocks ⟶ Desires / budgets ⟶ Adapted movements ⟶ New stocks

At step k, density $\rho_{i\ell}^k$ and continuous stock $n_{i\ell}^k = \Delta x \rho_{i\ell}^k$ describe section i, lane $\ell \in \{E, A, B\}$. These labels name current lanes: early-merged E traffic joins A's stock. The declared member (**D**) has $N = 10$ sections, $\Delta x = 0.1\,\text{km}$ and $\Delta t = 1/600\,\text{h}$ (6 s).

3.2. Core modeling assumptions. The task and model choices (**D**) are:

- **Cohort and advice.** Initialize a finite cohort inside the road, independently of advice; no later traffic enters. Choose one common $v \in [30, 60]$ km/h at the start and hold it constant.
- **Common lane physics.** Fix the geometry and use one common nominal triangular flow–density diagram: nominal free speed $u_0 = 60$, backward-wave speed magnitude $w = 20$ km/h, jam density $k_J = 150$ veh/km/lane and maximum capacity-drop fraction $\delta = 0.1$. Initial densities lie in $[0, k_J]$; mass is continuous.
- **Exit and accounting.** Trips end when traffic leaves through A or B; downstream congestion does not block the exit. Count every initial unit's time until it leaves, including traffic still queued at the lane end.
- **Service and updates.** All movements use old traffic and space and advance at most one section per step. Use equal positive priorities and no extra merging-headway factor. The grid satisfies $u_0\Delta t, w\Delta t \le \Delta x$.
- **Sensitivity.** Differentiate the complete objective where the active traffic rules remain unchanged; switching cases need separate treatment.

3.3. Sending and receiving capacity. Movement needs both traffic able to leave and room to accept it. The *sending rate* D is how much a cell can release per hour; the *receiving rate* R is how much the target lane can accept per hour from all competing streams. Both are in vehicle-equivalents per hour. Subraveti adopts Han for capacity drop [3]; we follow that inherited formulation (B) [4] with nominal/VSL caps (B) [5]. Our completion (**D**) keeps the full sending ceiling, without turning it off merely because the next cell becomes denser. This exact variant remains ours. FD identities are standard (S). The discharge ceiling $Q(x)$ equals nominal capacity C below critical density ρ_c, then falls linearly to $(1-\delta)C$ at jam density k_J:

$$C = \frac{u_0 w k_J}{u_0 + w}, \quad \rho_c = C/u_0, \quad Q(x) = \min\left\{C, C\left[1 - \delta\frac{x - \rho_c}{k_J - \rho_c}\right]\right\}, \tag{1}$$

$$D_{i\ell} = \min\{v\rho_{i\ell}, Q(\rho_{i-1,\ell})\}, \quad R(x,y) = \min\{Q(x), w(k_J - y), w(k_J - x) + w_d(x - y)\}. \tag{2}$$

Here x, y are the current upstream/downstream densities in the receiving lane. The three receiving bounds express discharge capacity, downstream space and Han's additional congestion-wave constraint. Its fixed recovery slope is $w_d = Q_J/(k_J - Q_J/u_0)$, $Q_J = (1-\delta)C$. Our lane-wise budget mapping (**A**) gives $d_{i\ell} = \Delta t D_{i\ell}$ and $r_{ij} = \Delta t R(\rho_{ij}, \rho_{i+1,j})$, amounts available in one step. All straight and diagonal arrivals to target j share that *one* r_{ij}. At the exit, $r_{Nj} = \Delta t R(\rho_{Nj}, 0)$, $j = A, B$, represents an unblocked downstream road; upstream boundary density is zero. Advice caps sending; receiving changes through the evolving densities, not directly through v.

3.4. Lane-change incentives. This component describes why traffic wants to move over before the lane ends. Subraveti's desire rule (B) [3] compares perceived densities K, with current/two-cell lookahead, weights 2:2:1 (B) [7]. The dimensionless multiplier I expresses the incentive for a particular lane change; $P_{i,\ell j}$ is the desired lateral proportion before shared traffic/service limits are applied:

$$P_{i,\ell j} = \left[\frac{I_{i,\ell j} K_{i\ell} - K_{ij}}{K_{i\ell} + K_{ij}}\right]_+, \qquad I = 1 + I^r + I^{kr} + I^{coop}, \qquad [x]_+ = \max(x, 0). \tag{3}$$

With $I = 1$, only a less crowded destination attracts traffic. A larger multiplier can motivate a change even toward a more crowded lane. The added terms represent route urgency, keep-right preference and cooperation. Within the last $D_r = 0.75$ km, E-to-A route bias (B) grows as $I^r = (1 - g_i/D_r)^3$, where $g_i = (N - i)\Delta x$; it is zero farther upstream. This adds urgency, not a fixed merge fraction or a hard start line: density incentives can cause earlier E-to-A movement. Member conventions (**D**) select mode/direction branches for keep-right/cooperation; free A can be encouraged toward B to make room. Precise mode and finite-end conventions are in the companion formulation; they add no separate traffic component.

3.5. Movement allocation and forced merge. Desire does not guarantee a lane change: requests must share the available traffic and receiving space. Before E ends, a failed attempt may still continue forward in E; at its end, traffic must merge or wait. Movement adaptation (**A**) [6] replaces native same-cell lateral/adjusted-forward coupling. Our composition (**D**) uses lane indices for each section: ℓ is the origin, j the destination. The desired lateral proportion $P_{\ell j}$ becomes an assigned fraction $b_{\ell j}$ of the origin's sending amount d_ℓ; $m_{\ell j}$ is the resulting request. Proportional normalization (**A**) preserves desires whose sum is at most one, otherwise scales them to fit one shared sending amount. The remainder goes straight. Two 70% lateral requests therefore become 50% each, rather than spending 140% of the same traffic:

$$b_{\ell j} = \frac{P_{\ell j}}{\max(1, \sum_{h \neq \ell} P_{\ell h})} \ (j \neq \ell), \quad b_{\ell\ell} = 1 - \sum_{j \neq \ell} b_{\ell j}, \quad m_{\ell j} = d_\ell b_{\ell j}. \tag{4}$$

We adapt R2's proportional receiving rule (**A**) [6] to the sender's budget; this placement is ours. The straight remainder (S) is accounting; request formation (B) follows [3, Eq. (14)]. R3/A3 sharing and forward fallback (**A**) [6] underpin the replacement. We extend its two-stream sharing to all target requests; Wright's priority formulation [8] provides supporting justification, without importing its general junction model.

For each target, choose the common service level λ_j so that the served amounts $z_{\ell j}$ fill its receiving amount r_j, or satisfy all requests if they are smaller:

$$z_{\ell j} = \min(m_{\ell j}, \lambda_j), \qquad \sum_{\ell} z_{\ell j} = \min\left(r_j, \sum_{\ell} m_{\ell j}\right). \tag{5}$$

Small requests are fully served and release unused entitlement to the others. Forward continuation (**A**) lets traffic not served in that allocation still advance in its own lane if space remains. This additional flow y_ℓ is limited by unused sending and unused same-lane receiving. If receiving space also prevents forward movement, the traffic waits in its current cell. Thus $f_{\ell j} = z_{\ell j} + \mathbf{1}_{\{j=\ell\}} y_\ell$ (**D**) is realized movement. An unsuccessful E-to-A attempt can therefore continue in E if space remains. Final-E rule (**D**): $b_{EA} = 1$, $y_E = 0$; E must use A's outlet or wait. It competes with A and any B-to-A request for the same outlet budget. Equal priorities are not a constant realized merge proportion; demand and receiving state determine the flow. At the choke, E uses part of A's exit capacity, so A's own traffic may have to wait. The resulting accumulation in A can reduce space for later arrivals and propagate congestion upstream. There is no separate speed-mismatch or acceleration-recovery meter. Our normalization and three-lane sequencing remain behavioral adaptations beyond the lane-end geometry.

Diagonal transport (**A**) advances one section when changing lanes. Apply old-state flows simultaneously using standard conservation (S):

$$n_{i\ell}^{k+1} = n_{i\ell}^{k} - \sum_{j} f_{i,\ell j}^{k} + \sum_{h} f_{i-1,h\ell}^{k}. \tag{6}$$

Set $f_0 = 0$; f_N counts traffic leaving the modeled road. Matched additions/subtractions conserve mass, while donor/receiver limits protect stock and storage.

3.6. Total system time and sensitivity. Minimize total system time J: all initial traffic's time until exit. Once remaining mass is below $\Delta x \rho_c$, all requests receive service and a unit in section i has remaining mean time $(N - i + 1)\Delta x / v$. Choose a fixed index K_* inside that regime. The weight $h_{i\ell} = N - i + 1$ counts the remaining cell advances from section i to the exit; $\mathbf{1}^\top n^k$ is the mass still on the road. The complete objective is

$$J(v) = \Delta t \sum_{k<K_*} \mathbf{1}^\top n^k + \frac{\Delta x}{v} h^\top n^{K_*}. \tag{7}$$

Population-time accounting is standard (S); the exact residual is derived here (**D**). The first term adds the time already spent by all unfinished traffic before the cutoff. The second adds all travel time still to come: each remaining stock is weighted by its distance to the exit and divided by advice speed. Traffic remaining at the cutoff is therefore counted rather than discarded. Write the complete update as $n^{k+1} = F(n^k, v)$. Between branch switches, $s^0 = 0$, $s^{k+1} = F_n^{[k]} s^k + F_v^{[k]}$ (S) propagates $s^k = \partial n^k / \partial v$ through budgets, desires, allocation and stock. Here F_n, F_v are the update's first derivatives with respect to state and advice. Then

$$J'(v) = \Delta t \sum_{k<K_*} \mathbf{1}^\top s^k + \frac{\Delta x}{v} h^\top s^{K_*} - \frac{\Delta x}{v^2} h^\top n^{K_*}. \tag{8}$$

This model's full derivative (**D**) uses standard chain rule (S). Terms measure earlier traffic changes, residual-stock changes and direct residual travel time. Hold a valid K_* fixed locally; at native mode or active-constraint switches a two-sided derivative can fail.

3.7. Validation design and rival models. The validation design (**D**) asks: does hiding A/B loading worsen advice, can one local reading suffice, and do gradients help choose advice? The planned comparisons keep the physical model and complete objective fixed:

1. Isolate information value. Compare an observer seeing E and A+B at each section with one seeing each lane. For paired states with identical reduced observations but different A/B loading, the reduced observer must choose one shared advice; the full observer may choose separately. Allow model-aware inference from admitted histories; decide before later discharge reveals the split.

2. Test a compact information rival. Add only final A-cell density to E and A+B. Its value, 60 or 90 veh/km/lane, identifies the reviewed initial state. Compare with full information to test whether one reading suffices; sufficiency beyond that pair is unproved.

3. Check gradients, then their usefulness. First check Section 3.6's complete-objective derivative against finite differences away from rule switches. Then, within each information view and with matched initialization and computational budget, compare gradient-guided advice with direct speed search and maximum advice. Search is a strong scalar-action reference, not a certified global optimum; maximum advice tests benefit over simply using the highest speed. This separates missing information from poor optimization. These checks are prospective, not reported results. The companion formulation specifies bias branches and finite-domain conventions.

References

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